

PCF 4.0 — Manage Supply Chain for Physical Products

HITL Maturity Assessment — All Processes

This document reports HITL (Human-in-the-Loop) maturity ratios for all of PCF 4.0 (Manage Supply Chain for Physical Products) — 4 process groups, 22 Level 3 processes, and 122 Level 4 activities in total — at five maturity levels from M1 (Undesigned) through M5 (Leading), with M3 marking HITL Design Intent.

At M3 (HITL Design Intent), this domain's aggregate human-involvement ratio is **50%**, down from a **72%** undesigned (M1) baseline — a **-22-point** AI convergence opportunity. Mixed-pattern L3 rows reflect an equal-weighted rollup of their real L4 children rather than a force-classified single pattern.

HITL Maturity Assessment

H% = Human involvement at each maturity level · M3 = HITL Design Intent · Full L1→L2→L3→L4 hierarchy

PCF	L1-Domain / L2-Process Group / L3-Process / L4-Activity	Pattern	M1	M2	M3 ★	M4	M5	Δ M1→M3
4.0	Manage Supply Chain for Physical Products · L1 DOMAIN AGGREGATE	—	72%	63%	50%	37%	28%	-22%
4.1	Plan for and align supply chain resources · L2 AVG	—	77%	68%	55%	42%	32%	-22%
4.1.1	Develop production and materials strategies		84%	76%	62%	49%	39%	-22%
4.1.1.1	Define manufacturing goals	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.1.2	Define labor and materials policies	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.1.3	Define outsourcing policies	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.1.4	Define capital expense policies	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.1.5	Define capacities	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.1.6	Define production network and supply constraints	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.1.7	Define production process	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.1.8	Define standard operating procedures	Document	72%	58%	40%	28%	18%	-32%
4.1.1.9	Define production workplace layout and infrastructure	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.2	Manage demand for products		78%	69%	56%	43%	33%	-22%
4.1.2.1	Develop baseline demand forecasts	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.2.2	Collaborate demand with customers	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.2.3	Develop demand consensus forecast	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.2.4	Determine available to promise	Transaction	60%	48%	35%	22%	12%	-25%
4.1.2.5	Monitor activity against demand forecast and revise forecast	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.2.6	Evaluate and revise demand forecasting approach	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.2.7	Measure demand forecast accuracy	Transaction	60%	48%	35%	22%	12%	-25%
4.1.3	Create materials plan		77%	68%	55%	42%	32%	-22%
4.1.3.1	Create unconstrained plan	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.3.2	Collaborate with supplier and contract manufacturers	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.3.3	Identify critical materials and supplier capacity	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.3.4	Monitor material specifications	Transaction	60%	48%	35%	22%	12%	-25%
4.1.3.5	Generate constrained plan	Transaction	60%	48%	35%	22%	12%	-25%
4.1.3.6	Define production balance and control	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.4	Create and manage master production schedule		68%	58%	45%	32%	22%	-23%
4.1.4.1	Model production network to enable simulation and optimization	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.4.2	Create master production schedule	Transaction	60%	48%	35%	22%	12%	-25%
4.1.4.3	Maintain master production schedule	Transaction	60%	48%	35%	22%	12%	-25%
4.1.5	Plan distribution requirements		62%	50%	38%	24%	14%	-24%
4.1.5.1	Maintain master data	Transaction	60%	48%	35%	22%	12%	-25%
4.1.5.2	Determine finished goods inventory requirements at destination	Transaction	60%	48%	35%	22%	12%	-25%

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4.1.5.3	Determine product storage facility requirements	Transaction	60%	48%	35%	22%	12%	-25%
4.1.5.4	Calculate requirements at destination	Transaction	60%	48%	35%	22%	12%	-25%
4.1.5.5	Calculate consolidation at source	Transaction	60%	48%	35%	22%	12%	-25%
4.1.5.6	Manage collaborative replenishment planning	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.5.7	Calculate and optimize destination dispatch plan	Transaction	60%	48%	35%	22%	12%	-25%
4.1.5.8	Manage dispatch plan attainment	Transaction	60%	48%	35%	22%	12%	-25%
4.1.5.9	Calculate and optimize destination load plans	Transaction	60%	48%	35%	22%	12%	-25%
4.1.5.10	Manage partner load plan	Transaction	60%	48%	35%	22%	12%	-25%
4.1.5.11	Manage cost of supply	Transaction	60%	48%	35%	22%	12%	-25%
4.1.5.12	Manage capacity utilization	Transaction	60%	48%	35%	22%	12%	-25%
4.1.6	Establish distribution planning constraints	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.6.1	Establish distribution center layout constraints	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.6.2	Establish inventory management constraints	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.6.3	Establish transportation management constraints	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.6.4	Establish storage management constraints	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.7	Review distribution planning policies		87%	81%	70%	59%	50%	-17%
4.1.7.1	Review distribution network	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.7.2	Establish sourcing relationships	Decision	92%	88%	80%	72%	65%	-12%
4.1.7.3	Establish dynamic deployment policies	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.8	Develop quality standards and procedures		76%	65%	48%	36%	26%	-28%
4.1.8.1	Establish quality targets	Knowledge	85%	78%	65%	52%	42%	-20%
4.1.8.2	Develop standard testing procedures	Document	72%	58%	40%	28%	18%	-32%
4.1.8.3	Communicate quality specifications	Document	72%	58%	40%	28%	18%	-32%
4.2	Procure materials and services · L2 AVG	—	77%	69%	57%	45%	35%	-20%
4.2.1	Provide sourcing governance	Knowledge	85%	78%	65%	52%	42%	-20%
4.2.1.1	Develop procurement plan	Knowledge	85%	78%	65%	52%	42%	-20%
4.2.1.2	Clarify purchasing requirements	Knowledge	85%	78%	65%	52%	42%	-20%
4.2.1.3	Establish materials management contingency plans	Knowledge	85%	78%	65%	52%	42%	-20%
4.2.1.4	Match needs to supply capabilities	Knowledge	85%	78%	65%	52%	42%	-20%
4.2.1.5	Analyze organization's spend profile	Knowledge	85%	78%	65%	52%	42%	-20%
4.2.1.6	Seek opportunities to improve efficiency and value	Knowledge	85%	78%	65%	52%	42%	-20%
4.2.1.7	Collaborate with suppliers to align with long term business goals	Knowledge	85%	78%	65%	52%	42%	-20%
4.2.1.8	Collaborate with suppliers to identify sourcing opportunities	Knowledge	85%	78%	65%	52%	42%	-20%
4.2.2	Develop category plan		87%	81%	70%	59%	50%	-17%
4.2.2.1	Analyze category/product positioning and performance	Knowledge	85%	78%	65%	52%	42%	-20%
4.2.2.2	Forecast demand for each category by market	Knowledge	85%	78%	65%	52%	42%	-20%
4.2.2.3	Assign revenue and margin targets to each category	Decision	92%	88%	80%	72%	65%	-12%
4.2.3	Select suppliers and develop/maintain contracts		76%	68%	58%	47%	38%	-18%
4.2.3.1	Select suppliers	Decision	92%	88%	80%	72%	65%	-12%
4.2.3.2	Certify and validate suppliers	Transaction	60%	48%	35%	22%	12%	-25%
4.2.3.3	Negotiate and establish contracts	Decision	92%	88%	80%	72%	65%	-12%

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PCF	L1-Domain / L2-Process Group / L3-Process / L4-Activity	Pattern	M1	M2	M3 ★	M4	M5	Δ M1→M3
4.2.3.4	Manage contracts	Transaction	60%	48%	35%	22%	12%	-25%
4.2.4	Order materials and services		70%	60%	48%	37%	28%	-22%
4.2.4.1	Process/Review requisitions	Transaction	60%	48%	35%	22%	12%	-25%
4.2.4.2	Approve requisitions	Decision	92%	88%	80%	72%	65%	-12%
4.2.4.3	Solicit supplier quotes	Transaction	60%	48%	35%	22%	12%	-25%
4.2.4.4	Create/Distribute purchase orders	Transaction	60%	48%	35%	22%	12%	-25%
4.2.4.5	Expedite orders and satisfy inquiries	Transaction	60%	48%	35%	22%	12%	-25%
4.2.4.6	Reconcile purchase orders	Transaction	60%	48%	35%	22%	12%	-25%
4.2.4.7	Research/Resolve order exceptions	Exception	95%	92%	82%	75%	68%	-13%
4.2.5	Manage suppliers		66%	56%	42%	30%	20%	-24%
4.2.5.1	Monitor/Manage supplier information	Transaction	60%	48%	35%	22%	12%	-25%
4.2.5.2	Prepare/Analyze procurement and supplier performance	Knowledge	85%	78%	65%	52%	42%	-20%
4.2.5.3	Support inventory and production processes	Transaction	60%	48%	35%	22%	12%	-25%
4.2.5.4	Monitor quality of product delivered	Transaction	60%	48%	35%	22%	12%	-25%
4.3	Produce/Assemble/Test product - L2 AVG	—	70%	60%	47%	35%	25%	-23%
4.3.1	Schedule production		64%	52%	39%	26%	16%	-25%
4.3.1.1	Model and simulate plant	Knowledge	85%	78%	65%	52%	42%	-20%
4.3.1.2	Generate line level plan	Transaction	60%	48%	35%	22%	12%	-25%
4.3.1.3	Generate detailed schedule	Transaction	60%	48%	35%	22%	12%	-25%
4.3.1.4	Schedule production orders and create lots	Transaction	60%	48%	35%	22%	12%	-25%
4.3.1.5	Plan for preventive maintenance	Transaction	60%	48%	35%	22%	12%	-25%
4.3.1.6	Request unplanned maintenance	Transaction	60%	48%	35%	22%	12%	-25%
4.3.1.7	Release production orders and create lots	Transaction	60%	48%	35%	22%	12%	-25%
4.3.2	Produce/Assemble product		74%	65%	53%	41%	31%	-21%
4.3.2.1	Manage raw material inventory	Transaction	60%	48%	35%	22%	12%	-25%
4.3.2.2	Execute detailed line schedule	Transaction	60%	48%	35%	22%	12%	-25%
4.3.2.3	Report maintenance issues	Transaction	60%	48%	35%	22%	12%	-25%
4.3.2.4	Rerun defective items	Exception	95%	92%	82%	75%	68%	-13%
4.3.2.5	Monitor and optimize production process	Knowledge	85%	78%	65%	52%	42%	-20%
4.3.2.6	Assess production performance	Knowledge	85%	78%	65%	52%	42%	-20%
4.3.3	Perform quality testing		70%	60%	48%	36%	26%	-22%
4.3.3.1	Calibrate test equipment	Transaction	60%	48%	35%	22%	12%	-25%
4.3.3.2	Perform testing using the standard testing procedure	Transaction	60%	48%	35%	22%	12%	-25%
4.3.3.3	Manage quality samples	Transaction	60%	48%	35%	22%	12%	-25%
4.3.3.4	Record test results	Transaction	60%	48%	35%	22%	12%	-25%
4.3.3.5	Track and analyze non-conformance trends	Knowledge	85%	78%	65%	52%	42%	-20%
4.3.3.6	Perform root cause analysis	Exception	95%	92%	82%	75%	68%	-13%
4.3.4	Maintain production records and manage lot traceability		72%	63%	50%	37%	27%	-22%
4.3.4.1	Determine lot numbering system	Knowledge	85%	78%	65%	52%	42%	-20%
4.3.4.2	Determine lot use	Transaction	60%	48%	35%	22%	12%	-25%
4.4	Manage logistics and warehousing - L2 AVG	—	65%	53%	40%	27%	17%	-25%
4.4.1	Provide logistics governance		83%	75%	61%	49%	39%	-22%
4.4.1.1	Translate customer service requirements into logistics requirements	Knowledge	85%	78%	65%	52%	42%	-20%
4.4.1.2	Design logistics network	Knowledge	85%	78%	65%	52%	42%	-20%
4.4.1.3	Communicate outsourcing needs	Document	72%	58%	40%	28%	18%	-32%
4.4.1.4	Develop and maintain delivery service policy	Knowledge	85%	78%	65%	52%	42%	-20%

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PCF	L1-Domain / L2-Process Group / L3-Process / L4-Activity	Pattern	M1	M2	M3 ★	M4	M5	Δ M1→M3
4.4.1.5	Optimize transportation schedules and costs	Knowledge	85%	78%	65%	52%	42%	-20%
4.4.1.6	Define key performance measures	Knowledge	85%	78%	65%	52%	42%	-20%
4.4.1.7	Define reverse logistics strategy	Knowledge	85%	78%	65%	52%	42%	-20%
4.4.2	Plan and manage inbound material flow	Transaction	60%	48%	35%	22%	12%	-25%
4.4.2.1	Plan inbound material receipts	Transaction	60%	48%	35%	22%	12%	-25%
4.4.2.2	Manage inbound material flow	Transaction	60%	48%	35%	22%	12%	-25%
4.4.2.3	Monitor inbound delivery performance	Transaction	60%	48%	35%	22%	12%	-25%
4.4.2.4	Manage flow of returned products	Transaction	60%	48%	35%	22%	12%	-25%
4.4.2.5	Manage disposition of returned products	Transaction	60%	48%	35%	22%	12%	-25%
4.4.3	Operate warehousing	Transaction	60%	48%	35%	22%	12%	-25%
4.4.3.1	Manage and track inventory deployment	Transaction	60%	48%	35%	22%	12%	-25%
4.4.3.2	Receive, inspect, and store inbound deliveries	Transaction	60%	48%	35%	22%	12%	-25%
4.4.3.3	Track product availability	Transaction	60%	48%	35%	22%	12%	-25%
4.4.3.4	Pick, pack, and ship product for delivery	Transaction	60%	48%	35%	22%	12%	-25%
4.4.3.5	Track inventory accuracy	Transaction	60%	48%	35%	22%	12%	-25%
4.4.3.6	Track third-party logistics storage and shipping performance	Transaction	60%	48%	35%	22%	12%	-25%
4.4.3.7	Manage physical finished goods inventory	Transaction	60%	48%	35%	22%	12%	-25%
4.4.3.8	Manage warehouse transfers	Transaction	60%	48%	35%	22%	12%	-25%
4.4.4	Operate outbound transportation	Transaction	60%	48%	35%	22%	12%	-25%
4.4.4.1	Plan, transport, and deliver outbound product	Transaction	60%	48%	35%	22%	12%	-25%
4.4.4.2	Track carrier delivery performance	Transaction	60%	48%	35%	22%	12%	-25%
4.4.4.3	Manage transportation fleet	Transaction	60%	48%	35%	22%	12%	-25%
4.4.4.4	Process and audit carrier invoices and documents	Transaction	60%	48%	35%	22%	12%	-25%
4.4.5	Deliver last mile delivery services	Transaction	60%	48%	35%	22%	12%	-25%
4.4.5.1	Confirm specific last mile delivery service requirements for individual customer	Transaction	60%	48%	35%	22%	12%	-25%
4.4.5.2	Identify and schedule resources to meet last mile delivery requirements	Transaction	60%	48%	35%	22%	12%	-25%
4.4.5.3	Provide last mile delivery service to specific customers	Transaction	60%	48%	35%	22%	12%	-25%
4.4.5.4	Ensure quality of last mile delivery	Transaction	60%	48%	35%	22%	12%	-25%

■ Decision
 ■ Knowledge
 ■ Document
 ■ Transaction
 ■ Exception

Full L1→L2→L3→L4 hierarchy · Mixed-pattern rows show an equal-weighted rollout of their real L4 children · © Timothy P. King & Claude (Anthropic) 2026